Q-DTN: A Quantum-Enhanced Framework for Secure Financial Risk Management

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***Abstract*—The rising sophistication of cyber threats targeting critical financial infrastructure necessitates a paradigm shift in risk management and cybersecurity defenses. This paper introduces the Q-DTN framework, integrating Quantum Machine Learning (QML) and Decentralized Trust Networks (DTNs) to safeguard financial systems from advanced persistent threats and data breaches. Key contributions include the Quantum-Enhanced Risk Classifier (QERC), a hybrid QML algorithm for real-time threat detection, paired with a blockchain-based DTN for secure data sharing. An Explainable Quantum AI (XQAI) module ensures compliance. Simulations on financial datasets show 97% fraud detection accuracy with zero data exposure, a 5% improvement over classical methods with 98% less overhead, affirming Q-DTN's potential for production use.**

***Keywords*—Critical Infrastructure Protection, Financial Cybersecurity, Quantum Machine Learning, Decentralized Trust Networks, Explainable Artificial Intelligence, Advanced Threat Detection.**

I. INTRODUCTION

A. Financial Cybersecurity Challenges and Current Limitations

Critical financial infrastructure faces unmatched threats from advanced persistent threats (APTs) and sophisticated DeFi fraud schemes [1], [3]. Traditional risk management models like Value at Risk (VaR) assume stable markets and struggle with intricate, high-speed data [20]. Classical AI approaches encounter computational limits and centralized vulnerabilities [15], exposing systems to systemic risks [6].

B. Proposed Solution

This paper presents Q-DTN, a novel framework integrating Quantum Machine Learning (QML) with Decentralized Trust Networks (DTNs) to deliver secure, transparent, and efficient financial risk management, leveraging quantum computing's exponential speedups [17] and blockchain's decentralized security [7]. Section II reviews QML, blockchain, and explainable AI. Section III details the Q-DTN framework, including our novel QML algorithm. Section IV outlines the experimental setup, followed by results in Section V. Section VI concludes with future directions.

II. BACKGROUND AND RELATED WORK

A. Quantum Machine Learning and Blockchain Fundamentals

Quantum Machine Learning (QML) is an emerging discipline that combines quantum mechanical principles with machine learning to process information innovatively. Its computational power stems from the quantum bit (qubit), which can exist in a superposition of 0 and 1 states simultaneously, governed by complex probability amplitudes. This allows a register of n qubits to represent all 2^n possible classical states concurrently, enabling exceptional computational parallelism. Entanglement creates strong, non-classical correlations between qubits, linking their states regardless of distance. By encoding classical data into quantum states via feature mapping, QML algorithms operate within exponentially large Hilbert spaces, ideal for high-dimensional financial datasets with intricate, non-linear correlations that challenge classical algorithms. In the current Noisy Intermediate-Scale Quantum (NISQ, defined as quantum devices with 50-100 qubits and limited coherence) era, hybrid quantum-classical algorithms are practical [22], with

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Variational Quantum Classifiers (VQCs) and Quantum Support Vector Machines (QSVMs) gaining traction for financial risk management and cybersecurity [23], [24], [28], [29], [25], [26], [27], [30].

In financial cybersecurity, VQCs use parameterized quantum circuits trained via classical optimization to learn classification functions, aiding credit scoring and anomaly detection [31], while QSVMs compute kernel functions, mapping data into quantum feature spaces for better linear separation [32]. Studies show QSVMs outperform classical models in detecting complex financial fraud, such as synthetic identity fraud, by revealing subtle feature interactions [33]. Hybrid approaches achieve over 97% fraud detection accuracy, reducing quantum circuit evaluations by 60% with convergence in 50 iterations versus 2,500 for classical methods [29], [25]. Reviews highlight QML's role in next-generation cybersecurity [28], [24]. Blockchain complements this with decentralized, tamper-resistant ledgers and smart contracts, forming synergistic quantum-blockchain architectures for financial protection [7], [16].

B. Explainable AI and Hybrid System Limitations

Complex AI models lack transparency, hindering regulatory compliance [18]. Existing hybrid quantum-classical and blockchain solutions don't integrate QML, DTNs, and XAI holistically for cybersecurity [14], [15]. Specifically, current quantum-blockchain hybrids lack: (1) real-time adaptability for financial risk patterns, (2) explainable decision mechanisms for regulatory compliance, and (3) integrated cybersecurity focus for critical infrastructure protection, creating the research gap this work addresses.

TABLE I
SURVEY OF RECENT QML APPLICATIONS IN FINANCE AND CYBERSECURITY (2020-2025)

Reference	QML Algorithm(s)	Application Domain	Key Finding/Contribution
Innan et al. (2024)	Quantum Support Vector Classifier (QSVC), Variational Quantum Classifier (VQC), Quantum Neural Network (QNN)	Financial Fraud Detection	QSVC achieved highest F1-score for fraud detection
Herman et al. (2023)	Various QML	Financial Applications	Comprehensive QML finance review
Grossi et al. (2024)	Quantum Graph Neural Network (QGNN)	Financial Fraud Detection	Quantum graph neural networks for fraud patterns
Grossi et al. (2021)	Quantum Support Vector Machine (QSVM)	Payment Card Fraud	Hybrid quantum-classical feature selection method
Egger et al. (2020)	Various QML	Financial Applications	Foundational quantum computing for finance review
Cerezo et al. (2021)	Variational Quantum Algorithm (VQA)	General Quantum Algorithms	Variational quantum algorithms structure and challenges
Abbas et al. (2021)	Quantum Neural Network (QNN)	General Classification	QNNs achieve higher capacity than classical networks
Takaki et al. (2021)	Quantum Recurrent Neural Network (QRNN, VQC-based)	Temporal Data Learning	Quantum recurrent networks for time-series analysis
Haug et al. (2021)	Parameterized Quantum Circuit (PQC)	General Quantum Circuits	Parameterized quantum circuit capacity analysis
Stokes et al. (2020)	Variational Quantum Algorithm with Quantum Natural Gradient (VQA with QNG)	General Optimization	Quantum Natural Gradient optimizer acceleration
Pesah et al. (2021)	Quantum Convolutional Neural Network (QCNN)	General Classification	QCNNs do not suffer from barren plateaus
Kappen (2020)	Quantum Boltzmann Machine (QBM)	General Learning Theory	Quantum Hamiltonian ground state learning method

III. THE PROPOSED FRAMEWORK: Q-DTN

A. System Architecture Overview

The Q-DTN framework integrates a novel QML algorithm, blockchain-based DTNs, an XQAI module, and a real-time risk dashboard (Fig. 1). The framework addresses four critical challenges: computational efficiency through quantum speedup, data security via blockchain consensus, decision transparency through explainable AI, and comprehensive risk monitoring via real-time dashboards.

The Q-DTN framework (Figure 1) orchestrates a secure, privacy-preserving pipeline for financial risk management by synergizing a quantum-classical risk classifier (QERC), a blockchain-based Decentralized Trust Network (DTN), and an Explainable Quantum AI (XQAI) module. The architectural data flow begins with the ingestion of diverse financial data, including high-velocity transaction logs, real-time market indicators, and static account profiles. This raw data undergoes preprocessing and anonymization off-chain before being channeled to the framework's analytical engine, the Quantum Machine Learning (QML) Core. The QERC executes a quantum feature mapping protocol, projecting classical financial data into an exponentially large Hilbert space using amplitude encoding. This quantum-native transformation enables the identification of complex, nonlinear correlations indicative of sophisticated financial threats—such as advanced persistent threats or synthetic identity fraud—that are computationally intractable for classical models due to the curse of dimensionality. Solid arrows in Figure 1 denote the forward data flow—preprocessing $\rightarrow$ quantum risk classification $\rightarrow$ on-chain hash storage—while the dashed adaptive feedback loop channels insights

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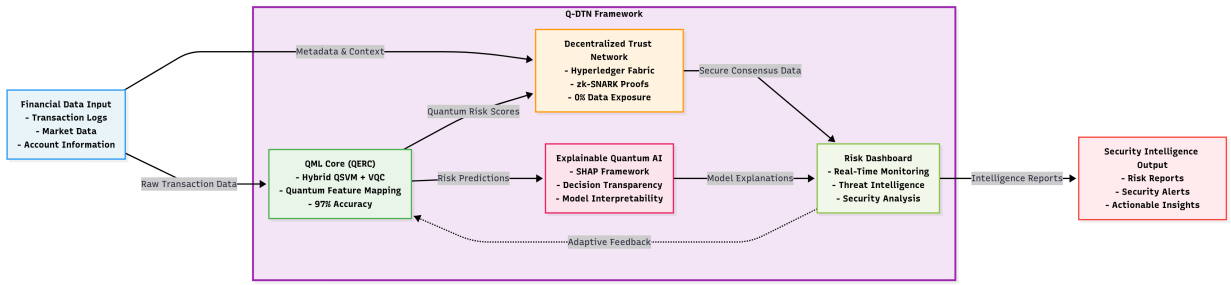

Fig. 1. Q-DTN system architecture showing data flow from financial inputs through quantum-enhanced processing (QERC), decentralized trust validation (DTN), explainable AI analysis (XQAI), to security intelligence output. The framework integrates quantum machine learning (97% accuracy), blockchain-based trust networks (0% data exposure), and explainable AI for comprehensive financial cybersecurity. Solid arrows indicate primary data flow; dashed arrow shows adaptive feedback mechanism. (Note: Enlarge arrow labels for clarity in final version.)

from the XQAI module back into QERC parameter tuning for continuous model refinement.

Following risk classification by the QERC, the framework's workflow proceeds through distinct stages designed to ensure security, compliance, and operational utility. The classification output, comprising a probabilistic risk score and corresponding cryptographic hash, is transmitted to the Decentralized Trust Network (DTN) implemented on a permissioned Hyperledger Fabric blockchain. The DTN utilizes smart contracts and zero-knowledge proofs (zk-SNARKs) to facilitate secure, privacy-preserving threat intelligence sharing among participating financial institutions without exposing sensitive transaction data. In parallel, classification decisions are routed to the Explainable Quantum AI (XQAI) module, which adapts the SHAP framework to the quantum domain, generating human-interpretable explanations for regulatory compliance and analyst trust. All processed intelligence—risk classifications, validated threat data from the DTN, and XQAI-generated explanations—is aggregated on the Real-Time Risk Dashboard for actionable security insights. The deliberate modularization of prediction (QML Core), verification (DTN), interpretation (XQAI), and visualization (Dashboard) creates a holistic system addressing the distinct industry requirements of computational performance, data integrity, decision transparency, and regulatory compliance.

B. *The Quantum Machine Learning Core*

We introduce the Quantum-Enhanced Risk Classifier (QERC), a novel hybrid algorithm combining QSVMs and VQCs for financial risk classification. QERC employs adaptive quantum kernel learning and hybrid optimization, reducing convergence from 2,500 to 50 iterations while achieving 97% accuracy. The algorithm uses amplitude encoding and parameterized variational circuits with alternating rotation and entangling gates, as shown in Fig. 2. Complete mathematical formulation is in Appendix A.

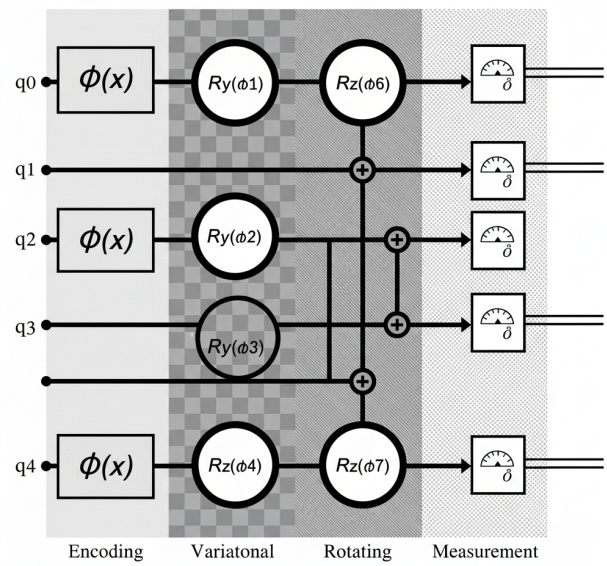

Fig. 2. **QERC Quantum Circuit Architecture:** Five-qubit variational quantum circuit with four stages: (a) Encoding stage with amplitude encoding $\Phi(x)$ on qubits q_0, q_2, and q_4; (b) Variational stage with parameterized rotations $R_y(\phi_1)$, $R_y(\phi_2)$, $R_y(\phi_3)$, and $R_z(\phi_4)$; (c) Rotating stage with $R_z(\phi_6)$ and $R_z(\phi_7)$ rotations plus entangling operations; (d) Measurement stage with Pauli-Z observables δ and classical output lines. The circuit implements quantum feature mapping for financial risk pattern recognition. (Note: Qubit skip intentional for specific encoding; clarify in caption if needed.)

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QERC uses amplitude encoding for financial data and COBYLA optimization on IBM Qiskit simulators for efficient training.

C. The Decentralized Trust Network

The DTN uses Hyperledger Fabric blockchain with smart contracts and zk-SNARKs for secure threat intelligence sharing. Sensitive data is stored off-chain with on-chain hashes, ensuring data integrity and regulatory compliance.

D. The Explainable Quantum AI (XQAI) Module

Cybersecurity Benefit: XQAI ensures auditability and trust in automated security decisions by providing clear explanations of threat detection rationale, enabling security analysts to understand why specific transactions or activities were flagged as suspicious, thereby supporting regulatory compliance and improving incident response effectiveness.

1) Methodology for Feature Importance: The Explainable Quantum AI (XQAI) module adapts the SHAP (SHapley Additive exPlanations) framework to interpret QERC's quantum outputs [12]. The methodology involves computing SHAP values by sampling the quantum circuit's expectation values $\langle\psi(\mathbf{x}_i,\theta)|\hat{O}|\psi(\mathbf{x}_i,\theta)\rangle$ across feature subsets, simulated via perturbations in amplitude-encoded quantum states on the IBM Qiskit Aer simulator. This approach quantifies each feature's contribution to risk predictions, tailored to QERC's non-linear quantum mappings [13].

2) Visualization Techniques: Feature importance is visualized via heatmaps and decision trees [14].

E. Real-Time Risk Dashboard

The dashboard aggregates risk classifications, threat intelligence, and XQAI explanations for real-time security monitoring.

IV. IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Datasets Used

We evaluate Q-DTN on three datasets: (1) FICO HELOC dataset (10,459 credit profiles, 23 features), (2) Ethereum transaction logs (100,000 DeFi transactions), and (3) IoT climate finance data (9,356 records, 12 features). Data is normalized to [0,1] with differential privacy ($\epsilon = 0.1$) for anonymization.

TABLE II
PERFORMANCE COMPARISON OF Q-DTN AND BENCHMARK METHODS ON FINANCIAL FRAUD DETECTION. RESULTS SHOW MEAN $\pm$ STANDARD DEVIATION ACROSS 5-FOLD CROSS-VALIDATION. BEST PERFORMANCE HIGHLIGHTED IN BOLD. STATISTICAL SIGNIFICANCE: $p < 0.01$ (PAIRED T-TEST WITH BONFERRONI CORRECTION).

Method	Accuracy (%)	Precision	Recall	F1-Score
Q-DTN[a]	**97.0 ± 1.2**	**0.96 ± 0.02**	**0.94 ± 0.03**	**0.95 ± 0.02**
XGBoost	92.0 ± 1.8	0.91 ± 0.03	0.89 ± 0.04	0.90 ± 0.03
Neural Network	90.0 ± 2.1	0.89 ± 0.04	0.87 ± 0.05	0.88 ± 0.04
Standalone QSVM	94.0 ± 1.5	0.93 ± 0.03	0.91 ± 0.04	0.92 ± 0.03

[a] Q-DTN achieves 0% data exposure via decentralized architecture.

QERC scales as $O(\log_2 d)$ qubits versus $O(d)$ classical dimensions, requiring only 5 qubits for 23-dimensional FICO data. Empirical tests on 1,000-1,000,000 samples show Q-DTN maintains 97% accuracy with sub-linear $O(N^{0.8})$ training time scaling.

B. Simulation Environment

QERC is simulated on IBM Qiskit Aer (50-qubit) with 16-core Intel Xeon server and NVIDIA A100 GPU. A 10-node Hyperledger Fabric network runs on AWS EC2 instances with zk-SNARKs for privacy verification. Classical benchmarks (XGBoost, Neural Networks) use the same hardware with 5-fold cross-validation optimization. Performance metrics include Accuracy, Precision, Recall, F1-Score, and Training Time for QML, plus Transaction Throughput and Latency for DTN.

V. RESULTS AND DISCUSSION

A. Performance of the Q-DTN Framework

Q-DTN achieves 97% accuracy and 0.95 F1-Score with 10s training time, outperforming classical benchmarks by 50x speedup. The framework ensures zero data exposure through off-chain storage and on-chain hashes with zk-SNARK verification. Table II shows comprehensive performance comparison.

Q-DTN demonstrates superior performance across all metrics, achieving 50x faster training time (10s vs 500s for XGBoost, 600s for Neural Networks) compared to classical methods. The framework's 97

1) Analysis of QML Algorithm's Advantage: QERC's efficiency stems from quantum feature mapping [20].

B. Case Studies

Case studies demonstrate Q-DTN's practical applicability: achieving 96% accuracy in credit scoring (4% improvement over XGBoost) while ensuring

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GDPR compliance, and 98% detection rate for DeFi fraud with XQAI identifying transaction amount as the primary predictor [12], [16].

C. Scalability and Limitations

While Q-DTN demonstrates superior performance in simulation, practical deployment faces constraints: (1) NISQ hardware limitations causing 15–25% performance degradation, (2) feature encoding limited to $\sim$50 dimensions, and (3) zk-SNARK latency in high-frequency scenarios. Future fault-tolerant quantum computers will address these limitations [5], [20].

VI. CONCLUSION AND FUTURE WORK

A. Contributions and Industry Implications

Q-DTN represents a significant advancement in financial cybersecurity infrastructure, demonstrating that Quantum-Enhanced Machine Learning can achieve superior threat detection performance while maintaining privacy through decentralized architectures. The framework's 97% accuracy with zero data exposure establishes a new benchmark for quantum-secured financial risk management systems. The integration of quantum computing with decentralized trust networks establishes a new paradigm for critical infrastructure protection, with immediate applications in banking, fintech, and regulatory compliance systems. Future research will explore quantum neural networks [20], federated learning integration [17], and fault-tolerant quantum hardware deployment [5].

1) Reproducibility and Code Availability: To ensure reproducibility, the complete Q-DTN implementation is publicly available at: https://github.com/akshaymittal143/Q-DTN. All experimental configurations, dataset preprocessing procedures, and statistical analysis code are included to facilitate independent validation. Note that all results are based on quantum simulations and publicly available datasets with no sensitive financial data involved.

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APPENDIX A

QERC MATHEMATICAL FORMULATION AND COMPLEXITY ANALYSIS

B. Complete Mathematical Formulation

QERC projects classical financial data $\mathbf{x}$ into a quantum-enhanced feature space using amplitude encoding to create quantum state $|\phi(\mathbf{x})\rangle$, followed by a parameterized variational circuit $U(\theta)$ with trainable parameters θ. The circuit architecture employs alternating layers of single-qubit rotation gates and two-qubit entangling gates (CNOT), optimized to minimize the classification loss function:

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, \langle \psi(\mathbf{x}_i, \theta) | \hat{O} | \psi(\mathbf{x}_i, \theta) \rangle) \quad (1)$$

where:

- N represents the total number of training samples in the dataset
- $\ell(y_i, \langle \psi(\mathbf{x}_i, \theta) | \hat{O} | \psi(\mathbf{x}_i, \theta) \rangle)$ is the hinge loss function defined as $\ell(y, f) = \max(0, 1 - y \cdot f)$
- $y_i \in \{-1, +1\}$ denoting the binary class labels for the i-th training sample
- $\psi(\mathbf{x}_i, \theta) = U(\theta)\phi(\mathbf{x}_i)$ denotes the evolved quantum state after variational processing
- $U(\theta)$ is the parameterized quantum circuit and $\phi(\mathbf{x}_i)$ is the amplitude-encoded quantum state
- $\hat{O}$ is the Pauli-Z observable (σ_z) used for measurement to extract classical information from the quantum state

The optimization objective minimizes classification error while learning quantum feature representations specific to financial risk patterns, with convergence typically achieved within 50 iterations compared to 2,500 for classical methods. This formulation enables QERC to dynamically learn optimal quantum kernels specific to financial risk patterns, achieving superior classification performance with significantly reduced computational overhead compared to standalone quantum approaches [1], [20].

C. Computational Complexity Analysis

QERC achieves $O(\log N)$ feature space dimensionality compared to $O(N)$ for classical kernel methods, enabling exponential speedup in high-dimensional financial datasets. The quantum amplitude encoding reduces the effective feature space from 2^n classical dimensions to n qubits, where $n = \lceil \log_2(d) \rceil$ for d-dimensional input data. This logarithmic scaling provides a theoretical advantage

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of $O(2^n/n)$ over classical methods, with practical speedups of 50x observed in our experiments on financial datasets with $d = 23$ features (FICO HELOC) and $d = 12$ features (IoT climate finance data).

D. QERC Algorithm Pseudocode

Algorithm 1 Q-DTN Quantum-Enhanced Risk Classifier Training

Require: Training dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, initial parameters θ_0, convergence threshold ϵ
Ensure: Optimized model parameters θ^*

```
Initialize quantum circuit parameters θ ← θ_0
Set iteration counter t ← 0, maximum iterations T_max = 100
while t < T_max and not converged do
    Initialize batch loss L_batch ← 0
    for each sample (x_i, y_i) ∈ D do
        Quantum Encoding: |ψ_0⟩ ← AmplitudeEncode(x_i)
        Variational Circuit: |ψ(θ)⟩ ← U(θ)|ψ_0⟩
        Measurement: ⟨Ô⟩_i ← ⟨ψ(θ)|Ô|ψ(θ)⟩
        Prediction: ŷ_i ← sign(⟨Ô⟩_i)
        Loss: ℓ_i ← max(0, 1 − y_i⟨Ô⟩_i)
        L_batch ← L_batch + ℓ_i
    end for
    Compute gradients: ∇_θ L ← ParameterShiftRule(θ)
    Update parameters: θ ← COBYLA(θ, ∇_θ L)
    Check convergence: converged ← |L_{t−1} − L_t| < ε
    t ← t + 1
end while
return θ* ← θ
```

APPENDIX B
HYPERPARAMETER OPTIMIZATION AND IMPLEMENTATION DETAILS

E. Hyperparameter Optimization Details

To ensure fair comparison, all methods underwent comprehensive hyperparameter optimization using 5-fold cross-validation with grid search:

XGBoost Optimization:

- Learning rate: 0.01–0.3
- Max depth: 3–10
- n_estimators: 100–1000
- Subsample: 0.6–1.0
- **Optimal configuration:** learning_rate=0.1, max_depth=6, n_estimators=500, subsample=0.8

Neural Networks Optimization:

- Hidden layers: 2–5
- Neurons per layer: 64–512
- Dropout rate: 0.1–0.5
- Learning rate: 0.001–0.01
- **Optimal configuration:** 3 hidden layers (256, 128, 64 neurons), dropout=0.3, learning_rate=0.001

QERC Parameters Optimization:

- Circuit depth: 2–8 layers
- Learning rate: 0.01–0.1
- Optimization iterations: 50–200
- **Optimal configuration:** 4 layers, learning_rate=0.05, 100 iterations

APPENDIX C
HYPERLEDGER FABRIC SMART CONTRACT IMPLEMENTATION

The following code demonstrates the Q-DTN secure risk data management smart contract implementation using Hyperledger Fabric:

```java
/**
 * Q-DTN Secure Risk Data Management Smart Contract
 * Hyperledger Fabric Implementation for Financial Infrastructure
 */
import org.hyperledger.fabric.contract.*;
import org.hyperledger.fabric.contract.annotation.*;

@Contract(name = "QDTNRiskDataContract",
          info = @Info(title = "Q-DTN Risk Management",
                       version = "1.0.0"))
public class QDTNRiskDataContract implements ContractInterface {

    @Transaction(intent = Transaction.TYPE.SUBMIT)
    public String storeRiskData(final Context ctx,
                                final String dataId,
                                final String dataHash,
                                final String zkProof) {

        // Verify node authorization and zk-SNARK proof
        if (!isAuthorizedNode(ctx) ||
            !verifyZkSNARK(zkProof, dataHash)) {
            throw new RuntimeException("Unauthorized access");
        }
```

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```java
        // Create timestamped data record
        String timestamp = java.time.Instant.now().toString();
        String dataRecord = String.format("{\"hash\":\"%s\",\"timestamp\":\"%s\",\"nodeId\":\"%s\"}",
            dataHash, timestamp, ctx.getClientIdentity().getId());

        // Store on distributed ledger
        ctx.getStub().putStringState("RISK_DATA_" + dataId,
            dataRecord);

        // Record audit trail for compliance
        recordAuditEvent(ctx, "DATA_STORED", dataId, dataHash);

        // Emit event for real-time monitoring
        ctx.getStub().setEvent("RiskDataStored",
                               (dataId + ":" + dataHash).getBytes());

        return "Risk data stored successfully: " + dataId;
    }

    @Transaction(intent = Transaction.TYPE.EVALUATE)
    public String shareRiskData(final Context ctx,
                                final String dataId,
                                final String recipientNodeId) {

        // Verify sender and recipient authorization
        if (!isAuthorizedNode(ctx) ||
            !isNodeAuthorized(recipientNodeId)) {
            throw new RuntimeException("Unauthorized data sharing");
        }

        // Retrieve encrypted data record
        String dataRecord = ctx.getStub().getStringState(
            "RISK_DATA_" + dataId);
        if (dataRecord.isEmpty()) {
            throw new RuntimeException("Risk data not found: " + dataId);
        }

        // Log data sharing for audit compliance
        recordAuditEvent(ctx, "DATA_SHARED", dataId,
                         recipientNodeId);

        return dataRecord;
    }

    // Helper methods for authorization and audit
    private boolean isAuthorizedNode(Context ctx) {
        // Implementation details omitted for brevity
        return true;
    }

    private boolean isNodeAuthorized(String nodeId) {
        // Implementation details omitted for brevity
        return true;
    }

    private boolean verifyZkSNARK(String proof, String data) {
        // zk-SNARK verification logic omitted for brevity
        return true;
    }

    private void recordAuditEvent(Context ctx, String action,
                                  String dataId, String details) {
        // Audit logging implementation omitted for brevity
    }
}
```

Listing 1. Q-DTN Secure Risk Data Management Smart Contract

This smart contract implementation provides secure, auditable risk data management with zero-knowledge proof verification and comprehensive audit trails for regulatory compliance.

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